Youyuan Zhang

Large Language Models · Multimodal Learning · Reinforcement Learning · AI Agents

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Education

Northeastern University (Project 985) China
Information Management and Information Systems *Sep. 2022 – Jun. 2026*
- **GPA:** 4.05 / 5.00 | **Overall Rank:** 1 / 78 (**Top 1.28%**) | **GPA Rank:** 3 / 78 (**Top 3.85%**)

Research Outputs

- **Toward Robust Personalized Alignment for LLMs: Mitigating Persona Drift in Multi-Turn Dialogue**
  Youyuan Zhang , Siyuan Li, Fangming Liu, Jing Li
  ***Accepted**, Findings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026 Findings)*
- **Modality-Decoupled Online Recursive Editing**
  Siyuan Li, **Youyuan Zhang** , Fangming Liu, Jing Li
  ***Accepted**, International Conference on Machine Learning (ICML 2026)*
- **SAIL: Separating User Signal from Task Context for Cold-Start GUI Agents**
  Youyuan Zhang , Siyuan Li, Jing Li
  ***Submitted**, International Conference on Learning Representations (ICLR 2027)*
- **Multimodal Knowledge Edit-Scoped Generalization for Online Recursive MLLM Editing**
  Siyuan Li, **Youyuan Zhang** , Ruitong Liu, Junxi Wang, Jing Li
  ***Submission**, IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI)*

Research Experience

Tsinghua University — Research Intern, Supervisor: Prof. Xinlei Chen *Apr. 2025 – Present*
- Developed a physics-constrained conditional diffusion model for probabilistic wildfire spread forecasting by integrating a pretrained physics simulator into denoising, achieving **76.82% AUPRC**.

Sony (China) Co., Ltd. — Algorithm Research Intern *Feb. 2026 – May 2026*
- Built an **8B VLM** post-training and inference pipeline with **180K SFT** and **42K preference/RL samples** using DeepSpeed, Megatron-LM, and vLLM; improved MathVista from **68.2% to 72.1%**, MMMU-STEM from **58.6% to 61.8%**, and evaluation throughput by **2.7**×.

National-Level Undergraduate Innovation & Entrepreneurship Project — Project Lead *No. 202419145004*
- Developed a deep-learning crop seedling and weed detector; introduced adaptive convolution with dynamically adjusted kernel-sampling paths for slender and curved targets, improving **AP$_S$ by 6%** while enabling stable hardware deployment; the project received an **Excellent** completion rating and yielded **7 national-level** and **10 provincial-level awards**.

Selected Awards & Honors

- **National Scholarship**, 2023
- **National Encouragement Scholarship**, 2024, 2025
- **Top 10 Innovation and Entrepreneurship Star**(**Top 0.083%**), **Highest Innovation Honor, University-wide**
- **National First Prize**, National College Students Innovation & Entrepreneurship Annual Meeting, Most Popular Project
- **National Second Prize**, China International College Students' Innovation Competition
- **National Second Prize**, Challenge Cup China College Students' Entrepreneurship Competition, **Core Member**
- **National Second Prize**, Challenge Cup Black Technology Special Track, **Team Leader**
- **Provincial Special Prize**, Challenge Cup Academic Science & Technology Competition
- **Cumulative: 8 National-Level Awards, 16 Provincial-Level Awards**

Technical Skills

LLM / VLM Post-Training: SFT, GRPO, Preference / RL Data Construction, Multimodal Reasoning Data Synthesis
Distributed Training & Serving: DeepSpeed, Megatron-LM, vLLM, ZeRO, Tensor Parallelism, Pipeline Parallelism, Mixed Precision
Research: Large Language Models, Multimodal Learning, Reinforcement Learning, AI Agents, GUI Agents, Online Knowledge Editing, Diffusion Models, Bayesian Inference
Models / Architectures: BLIP-2, LLaVA, FLUX Klein 4B, EDM